

RETAIL SALES ANALYTICS

A Power BI Business Intelligence Case Study

End-to-End Analytics Engagement: Data Modeling, DAX Development, and Executive Dashboard Design

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Technology Stack: Power BI • DAX • Power Query • SQL • Python (Pandas, NumPy, Matplotlib, Seaborn) • Star Schema Data Modeling

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1. Executive Summary

The Retail Sales Analytics Dashboard was developed to transform transactional retail data into actionable business intelligence using Microsoft Power BI. Organizations often collect large volumes of operational data but struggle to derive meaningful insight from it due to fragmented reporting and manual analysis. This engagement addresses that challenge by designing an interactive executive dashboard that enables decision-makers to monitor business performance through dynamic visualizations and centralized KPI tracking.

The solution consolidates customer, product, order, and sales data into a unified reporting platform, allowing stakeholders to analyze sales trends, customer behavior, product performance, regional sales, and payment preferences from a single source of truth.

Snapshot of Reported Performance

1.72bn	5,000	500	45K	344.81K
Total Sales	Total Orders	Total Customers	Quantity Sold	Avg. Order Value

This document presents the complete engagement: the business problem, requirements, data architecture, transformation logic, DAX measure library, dashboard design rationale, and the resulting business recommendations. It is written in the format of a client-facing analytics deliverable, reflecting the standard a Business Intelligence Developer would produce for a stakeholder audience.

Project Deliverables

- Interactive Power BI Dashboard
- Star Schema data model
- DAX measure library
- Executive-ready dashboard views
- Business insights report
- Technical design documentation
- Consolidated data dictionary
- Business requirements documentation

2. Business Problem Statement

2.1 Background

Retail organizations generate thousands of transactions daily across multiple product categories, customers, payment methods, and geographic locations. Without a centralized reporting system, decision-makers often rely on manual spreadsheets, which leads to inconsistent reporting, delayed insight, and inefficient operations.

This engagement addresses these challenges by creating an interactive Power BI dashboard that consolidates sales information into a single, governed source of truth.

2.2 Existing Challenges

- Sales reports were generated manually and inconsistently across teams.
- Data was distributed across multiple disconnected tables and files.
- No centralized KPI monitoring existed for leadership.
- Sales trends were difficult to identify without manual pivoting.
- Customer distribution across states/regions was not visible at a glance.
- Regional performance could not be compared efficiently.
- Product category performance lacked visibility for merchandising decisions.
- Executives spent excessive time preparing static reports instead of acting on them.

2.3 Proposed Solution

- Interactive executive dashboard with drill-down capability
- Automated, centralized KPI calculations via DAX
- Dynamic filtering by year, quarter, month, category, and payment mode
- Regional sales analysis and comparison
- Category-wise performance tracking
- Customer distribution analysis by state
- Monthly sales trend monitoring
- Payment mode analysis
- Business-ready, presentation-quality visualizations

2.4 Cost of Inaction

Continuing with manual, spreadsheet-based reporting carries a compounding cost. Each reporting cycle consumes analyst hours that could otherwise go toward analysis rather than data assembly; inconsistent report versions erode stakeholder trust in the numbers; and slow turnaround means decisions are made on data that is already a cycle old by the time it reaches leadership. A centralized, self-service dashboard converts that recurring manual effort into a one-time modeling investment, after which insight is available on demand rather than on a reporting schedule.

3. Business Requirements Document (BRD)

3.1 Business Goal

To develop an interactive Business Intelligence dashboard capable of monitoring sales performance, customer analytics, and operational KPIs through a centralized reporting platform accessible to non-technical business stakeholders.

3.2 Stakeholders & Responsibilities

Stakeholder	Responsibility	Primary Interest
CEO	Business overview	Total sales, growth trend, overall health
Sales Manager	Sales monitoring	Regional and category performance
Marketing Manager	Customer trends	Customer distribution, acquisition patterns
Finance Team	Revenue analysis	Total sales, average order value, cost/discount impact
Operations Team	Product performance	Quantity sold, category-level demand

3.3 Key Performance Indicators (KPIs)

KPI	Description	Owner
Total Sales	Overall revenue generated (net of discounts)	Finance / CEO
Total Orders	Number of completed, unique orders	Sales Manager
Total Customers	Active, unique customer count	Marketing Manager
Quantity Sold	Total units sold across all products	Operations Team
Average Order Value	Revenue generated per order	Finance / Sales Manager

3.4 Scope

In scope: consolidation of customer, order, order-line, product, and calendar data into a single analytical model; development of a centralized DAX measure library; design of a single-page executive dashboard with slicers for Year, Quarter, Month, Category, and Payment Mode; and delivery of supporting documentation.

Out of scope: predictive/forecasting models, row-level security configuration, automated data refresh scheduling, and integration with external CRM/ERP systems. These are noted in Section 11 as recommended next-phase enhancements.

3.5 Assumptions & Constraints

- Source data is provided as flat, exportable tables (no live database connection in this phase).
- Currency and units are assumed consistent across all records post-cleaning.
- Reporting is monthly/quarterly cadence, not real-time streaming.
- Dashboard is designed for desktop and tablet viewing (Power BI Service).

3.6 Project Phases

Phase	Activities	Output
1. Discovery	Stakeholder interviews, KPI definition, scope agreement	Business Requirements Document
2. Data Preparation	Profiling in Python/SQL, Power Query cleansing	Validated, model-ready tables
3. Modeling	Star schema design, relationship configuration	Data model with documented cardinality
4. Measure Development	DAX measure library build and validation	Centralized KPI Measures table
5. Dashboard Design	Layout, slicers, visual selection, F-pattern layout	Executive dashboard (Page 1)
6. Review & Handoff	Stakeholder walkthrough, documentation package	This case study and supporting docs

4. Data Understanding & Data Dictionary

4.1 Dataset Overview

The Retail Sales Analytics solution integrates multiple business entities into a centralized analytical model. The dataset consists of six interconnected tables representing customers, orders, products, sales transactions, and time dimensions, structured using a Star Schema to ensure optimal query performance and scalability.

Table Name	Type	Purpose
Customers	Dimension	Customer demographics and geography
Orders	Dimension / Header	Order-level information (date, region, payment)
Order_Details	Fact	Transaction-level sales data
Products	Dimension	Product catalog and cost/price master
Calendar	Dimension	Date dimension for time intelligence
KPI Measures	Measure table	Centralized DAX business calculations

4.2 Customers Table

Business Purpose

Stores customer demographic information used for segmentation and regional analysis.

Column	Description
CustomerID	Unique identifier for each customer
CustomerName	Full customer name
Age	Customer age
Gender	Male / Female classification
City	Customer city
State	Customer state
JoinDate	Customer registration date

Business Value

- Customer segmentation
- Regional customer analysis
- Demographic reporting

4.3 Orders Table

Contains order-level information used to answer regional and payment-preference questions.

Column	Description
OrderID	Primary key
CustomerID	Foreign key — links to Customers
OrderDate	Purchase date
Region	Sales region
PaymentMode	Mode of payment used

Business Questions Answered

- Which region performs best?
- Which payment method is most preferred?
- How many orders are placed, and over what period?

4.4 Order_Details Table

Stores every product sold within every order — the fact table of the model.

Column	Description
OrderDetailID	Unique transaction identifier
OrderID	Foreign key — links to Orders
ProductID	Foreign key — links to Products
Quantity	Units sold
UnitPrice	Selling price per unit
Cost Price	Purchase cost per unit
Gross Sales	Sales value before discount
Discount	Discount percentage applied
Discount Amount	Discount value in currency
Net Sales	Final sales amount after discount

4.5 Products Table

Column	Description
ProductID	Product identifier

Column	Description
ProductName	Product name
Category	Product category
SubCategory	Product subcategory
UnitPrice	Selling price
CostPrice	Product cost

4.6 Calendar Table

The Calendar table enables Time Intelligence calculations and consistent filtering across the model. It was generated using DAX date functions rather than imported, ensuring a contiguous, gap-free date range aligned to the fiscal reporting calendar.

Column	Purpose
Date	Complete date — the join key to Orders[OrderDate]
Day	Day of month
Month No	Numeric month, used for chronological sorting
Quarter	Calendar quarter
Year	Calendar year

5. Data Cleaning & Transformation

5.1 Objective

Before building the dashboard, the raw data was validated and transformed to ensure accuracy, consistency, and analytical readiness. The ETL (Extract, Transform, Load) process was performed using Microsoft Power Query Editor.

5.2 Data Quality Assessment

Validation Check	Status
Missing Values	Reviewed
Duplicate Records	Checked
Invalid Data Types	Corrected
Blank Fields	Removed
Data Consistency	Verified
Relationship Integrity	Validated

5.3 Power Query Transformations Applied

Customers Table

- Verified CustomerID uniqueness
- Standardized State and City values (casing, whitespace)
- Converted JoinDate to Date format

Orders Table

- Converted OrderDate to Date type
- Verified OrderID uniqueness
- Standardized PaymentMode values

Order_Details Table

- Verified Quantity values (removed nulls/negative entries)
- Standardized Unit Price and Cost Price as Decimal type
- Validated Discount calculations against Gross and Net Sales

Products Table

- Removed redundant spaces from text fields
- Standardized Category names
- Verified ProductID uniqueness

Calendar Table

- Generated using DAX to guarantee full date coverage
- Derived Year, Quarter, Month Number, and Day fields for Time Intelligence

5.4 Why Data Cleaning Matters

Clean, validated data is the foundation every downstream KPI depends on. Poor data quality silently corrupts DAX aggregations, misleads stakeholders, and erodes trust in the dashboard. Cleaning at the Power Query layer — rather than patching with DAX workarounds — keeps the model performant and the logic auditable.

- Dashboard accuracy
- KPI reliability
- Report performance
- Business decision-making confidence
- Model consistency across refreshes

6. Data Modeling & Star Schema Architecture

6.1 Why Star Schema

A Star Schema architecture was implemented to simplify analytical queries and maximize reporting performance. It is the preferred modeling approach in Power BI because it:

- Improves report and visual rendering performance
- Simplifies DAX calculations by keeping context propagation predictable
- Reduces ambiguity in relationships between tables
- Enhances filter propagation from dimensions to facts
- Supports scalable business reporting as data volume grows

6.2 Relationship Summary

From	To	Cardinality	Cross-Filter Direction
Customers.CustomerID	Orders.CustomerID	One-to-Many	Single
Orders.OrderID	Order_Details.OrderID	One-to-Many	Single
Products.ProductID	Order_Details.ProductID	One-to-Many	Single
Calendar.Date	Orders.OrderDate	One-to-Many	Single

6.3 Conceptual Data Flow

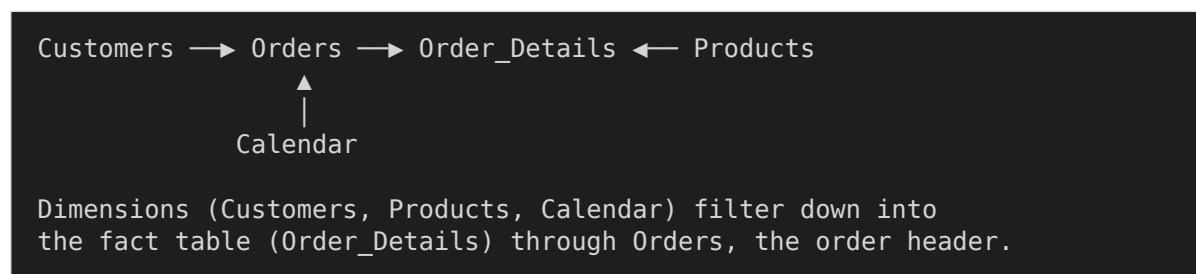


Figure 6.1 — Star schema relationship flow (dimensions surrounding the central fact table).

6.4 Design Rationale

This model separates transactional data (Order_Details) from descriptive dimensions (Customers, Products, Calendar), which minimizes redundancy and ensures business metrics remain consistent across every visual. Single-direction filtering was retained throughout to avoid ambiguous filter paths and circular dependency errors — a common pitfall in Power BI models that mix bidirectional relationships with many-to-many joins.

The KPI Measures table is an empty, disconnected table used purely to host the centralized DAX Measure Library, which is standard best practice: it keeps calculations separate from raw data columns, improves discoverability in the field list, and simplifies long-term maintenance.

6.5 Alternative Approaches Considered

A snowflake schema (further normalizing Products or Customers into sub-dimensions) was considered and rejected for this use case. With only six tables and no repeating attribute groups, the added join complexity of a snowflake design would have increased DAX complexity and slightly degraded query performance without a corresponding analytical benefit. Star schema was chosen deliberately, not by default.

A single flattened table (denormalizing everything into one wide table) was also considered. It was rejected because it would duplicate customer and product attributes across every transaction row, inflating model size and making many-to-one relationships like 'one customer, many orders' harder to express correctly in DAX.

7. DAX Measure Development

7.1 Introduction

DAX (Data Analysis Expressions) was used to create reusable business metrics that dynamically respond to user selections and report filters. All calculations were centralized in a dedicated KPI Measures table for maintainability and performance.

7.2 Measure 1 — Total Sales

Business Objective

Calculate total revenue generated across all completed sales transactions, net of discounts.

```
Total Sales =  
SUM(Order_Details[Net Sales])
```

Business Logic

Aggregates the Net Sales column and automatically responds to Year, Quarter, Category, Region, and Payment Mode filters via the model's relationships.

Used In

- KPI Card
- Monthly Sales Trend
- Sales by Category
- Sales by Region

7.3 Measure 2 — Total Orders

```
Total Orders =  
DISTINCTCOUNT(Orders[OrderID])
```

Purpose

Calculates the total number of unique orders placed — DISTINCTCOUNT rather than COUNT to avoid inflating the metric if the grain were ever duplicated upstream.

Business Use

- Order tracking
- Sales volume analysis
- Customer purchasing behavior

7.4 Measure 3 — Total Customers

```
Total Customers =  
DISTINCTCOUNT(Customers[CustomerID])
```

Purpose

Measures the total active customer base referenced by the current filter context.

Business Impact

Helps evaluate market reach and customer growth over time and across regions.

7.5 Measure 4 — Total Quantity Sold

```
Total Quantity Sold =  
SUM(Order_Details[Quantity])
```

Purpose

Calculates the total number of units sold.

Business Value

Useful for inventory planning and demand forecasting at the category and product level.

7.6 Measure 5 — Average Order Value

```
Average Order Value =  
DIVIDE([Total Sales], [Total Orders])
```

Business Interpretation

Indicates the average revenue generated per customer order. DIVIDE is used instead of the '/' operator to gracefully return BLANK() rather than an error when Total Orders is zero — important when slicers filter down to an empty context.

Management Use

Monitors purchasing behavior and evaluates the effectiveness of pricing and promotional strategies.

7.7 DAX Design Principles Applied

- Every measure lives in a single, centralized KPI Measures table rather than scattered across source tables.
- DIVIDE() is used for all ratio calculations to handle divide-by-zero safely.
- DISTINCTCOUNT is used wherever a table could contain duplicate grain, rather than assuming COUNT is safe.
- Measures are named in plain business language (Total Sales, not SumOfNetSales) so they read naturally in report tooltips and the field list.

7.8 Recommended Enhancements (Next Phase)

The following measures were identified as high-value extensions but sit outside the current phase scope — noted here to demonstrate forward planning:

```
Sales YTD = TOTALYTD([Total Sales], Calendar[Date])  
Sales Prior Year = CALCULATE([Total Sales],  
SAMEPERIODLASTYEAR(Calendar[Date]))
```

```
Sales YoY % = DIVIDE([Total Sales] - [Sales Prior Year], [Sales Prior Year])
```

These would require the Calendar table to be marked as an official Date table (Mark as Date Table) to support Time Intelligence functions correctly.

8. Dashboard Design Strategy

8.1 Design Philosophy

The dashboard was designed with a focus on executive usability, ensuring decision-makers can identify business trends within seconds. The layout follows the F-Pattern reading model, a UX principle commonly used in business intelligence dashboards: high-priority metrics sit in the upper-left region, while analytical visuals occupy the central and lower sections.

8.2 Dashboard Sections

KPI Summary

Provides an immediate overview of business performance: Total Sales, Total Orders, Total Customers, Quantity Sold, and Average Order Value, displayed as high-contrast cards along the top and left rail.

Interactive Filters

Users can dynamically filter the dashboard by Year, Quarter, Month, Category, and Payment Mode. These slicers allow real-time exploration without modifying the underlying data model.

Trend Analysis

The Monthly Sales Trend visual highlights changes in sales performance over time, enabling users to identify seasonal fluctuations and longer-term trends.

Regional Analysis

Sales by Region compares business performance across geographic areas, helping management identify high-performing and underperforming markets.

Product Analysis

Category-based visuals reveal each product category's contribution to overall revenue, supporting merchandising and inventory decisions.

8.3 Visual Interactivity

All visuals are fully interactive and cross-filtered. Selecting a value in Sales by Category, Sales by Region, Payment Mode, or Customer Distribution updates the entire dashboard automatically, ensuring consistent analysis across every KPI and chart.

9. Dashboard Walkthrough

9.1 KPI Cards

KPI Card	Purpose	Business Insight
Total Sales	Displays total revenue across all completed transactions	Primary financial KPI; quick overview of overall performance
Total Orders	Tracks total customer orders processed	Evaluates sales activity and operational workload
Total Customers	Displays total unique customer base	Supports acquisition and retention analysis
Total Quantity Sold	Measures total units sold	Useful for demand forecasting and inventory management
Average Order Value	Average revenue per order	Higher values indicate stronger spending and effective pricing

9.2 Core Visuals

Visual	Chart Type	Business Question Answered
Monthly Sales Trend	Line chart	How is revenue trending month over month?
Sales by Region	Column chart	Which regions drive the most revenue?
Category-wise Status	Donut chart	Which categories contribute most to overall sales?
Customer Distribution by State	Horizontal bar	Where is the customer base concentrated geographically?
Total Sales by Category	Horizontal bar	How does revenue break down across product categories?
Sales by Payment Mode	Donut chart	Which payment methods are customers using?

9.3 Layout Rationale

The left rail anchors branding and global filters (PaymentMode, Month No, Category, Quarter, Year) so a user's filtering intent is visible at all times. The top band holds the five KPI cards for at-a-glance orientation. The right two-thirds of the canvas is reserved for trend, regional, category, and customer-distribution visuals — arranged so that time-based analysis (trend) sits above comparative analysis (region, category, customers), matching how executives typically read a report: first 'how are we doing over time,' then 'where is that coming from.'

9.4 Guided Walkthrough Narrative

This section narrates the dashboard the way it would be presented in a stakeholder demo, moving left to right and top to bottom in the order an executive naturally scans the page.

Step 1 — Orientation (Left Rail)

The viewer starts at the branded left rail, which anchors the report identity and hosts every global slicer: Payment Mode, Month, Category, Quarter, and Year. Because filters live in one consistent location, a user never has to hunt for 'what is this view currently filtered to.'

Step 2 — Headline KPIs

Moving right, the five KPI cards (Average Order Value, Quantity Sold, Total Sales, Total Orders, Total Customers) give an immediate performance snapshot before any deeper analysis begins — answering 'how are we doing' in under five seconds.

Step 3 — Trend Context

The Monthly Sales Trend line chart, positioned top-right, answers the natural follow-up question: 'is that snapshot improving or declining over time?' Placing it immediately after the KPIs — rather than at the bottom of the page — respects the F-pattern reading order.

Step 4 — Where and What

The lower two rows answer 'where is this performance coming from': Sales by Region and Category-wise Status on the left-center, Customer Distribution by State and Total Sales by Category on the right, and Sales by Payment Mode along the bottom. Together they let a viewer decompose the headline number by geography, product, customer base, and payment channel without leaving the page.

Step 5 — Drill Down

Any single click — a region bar, a category slice, a payment-mode segment — cross-filters every other visual instantly, letting a stakeholder self-serve an answer to 'what does Furniture look like in the West region for UPI payments' without writing a query or asking an analyst.

10. Business Insights & Recommendations

10.1 Observed Performance Patterns

Based on the dashboard's current reporting period, the following patterns emerge from the data:

- Furniture leads category contribution at roughly 26.7% of total sales, followed by Grocery (~24.2%) and Clothing (~22.1%), indicating a relatively balanced category mix rather than dependence on a single product line.
- Payment mode is evenly distributed across Net Banking, Credit Card, Cash, Debit Card, and UPI (each between ~18-21% of transactions), suggesting no single payment channel dominates customer preference.
- Maharashtra, Delhi, and Tamil Nadu lead in customer count, indicating these states are the strongest markets for retention and loyalty programs.
- Monthly sales show visible month-to-month volatility (ranging roughly between 138M and 152M), which is a signal worth investigating for seasonality or promotional-cycle effects.
- Regional sales by column chart shows uneven distribution across regions, with a clear leading region and a comparatively underperforming one.

10.2 Recommendations

Observation	Recommendation	Expected Impact
Even payment-mode split	Avoid over-investing in a single payment gateway; ensure all five channels remain frictionless at checkout	Protects conversion across the full customer base
Furniture & Grocery lead category sales	Prioritize inventory and marketing spend toward these categories while testing promotions to lift Beauty and Sports	Improved category-level ROI on marketing spend
Concentration of customers in a few states	Design state-specific retention campaigns for Maharashtra, Delhi, and Tamil Nadu; run acquisition campaigns in lower-count states	Balanced growth across geographies, reduced regional risk
Month-to-month sales volatility	Layer in Time Intelligence measures (YTD, YoY) to distinguish seasonality from genuine decline	Better-informed forecasting and inventory planning
Underperforming region identified	Conduct a regional root-cause review (pricing, logistics, assortment) before reallocating budget	Targeted intervention instead of blanket budget cuts

10.3 Suggested Next-Phase Analytics

- Add Time Intelligence measures (YTD, MTD, YoY%) once the Calendar table is marked as an official date table.
- Introduce a customer RFM (Recency, Frequency, Monetary) segmentation to support targeted marketing.
- Add a cohort or repeat-purchase-rate view to distinguish acquisition-driven growth from retention-driven growth.

- Automate refresh scheduling once the data source moves from static extracts to a live database connection.

11. Technical Design Summary

11.1 Technology Stack

Layer	Tool(s)	Purpose
Data Preparation	Excel, Python (Pandas, NumPy)	Initial data profiling and cleansing
Exploratory Analysis	Google Colab, Matplotlib, Seaborn	Pattern discovery ahead of dashboard build
Data Querying	SQL	Structured extraction and validation queries
ETL / Transformation	Power Query	In-Power BI cleansing, type correction, standardization
Data Modeling	Power BI Desktop (Star Schema)	Relationship design and model optimization
Business Logic	DAX	Centralized measure library
Presentation Layer	Power BI Desktop / Service	Executive dashboard and distribution

11.2 Architecture Principles

- Single source of truth: all KPIs are derived from the same underlying fact table, eliminating conflicting numbers across visuals.
- Separation of concerns: raw data (Power Query), business logic (DAX), and presentation (report canvas) are kept distinct and independently maintainable.
- Performance-first modeling: Star Schema with single-direction relationships keeps query folding and visual rendering fast as data volume scales.
- Documentation-as-deliverable: every measure and transformation is documented so the model can be handed off or audited by another developer.

11.3 Data Governance Considerations

Although full governance tooling sits outside this phase's scope, the model was built with governance in mind: a single certified dataset, one measure per business definition (no duplicate 'Total Sales' logic hiding in a different table), and column-level documentation for every field, so a governance review has a clear starting point rather than a blank slate.

11.4 Performance Considerations

Power BI's VertiPaq engine compresses columnar data most efficiently when cardinality is low and relationships are simple — both of which this star schema was designed around. Practical steps taken to protect performance as data volume grows:

- Removed unused columns from each table before loading, since every loaded column consumes memory whether or not it is used in a visual.
- Kept all relationships single-direction to avoid the query-plan overhead of bidirectional filtering.
- Centralized measures avoid redundant calculation logic being evaluated in multiple places.
- Categorical columns (Category, Region, PaymentMode) were kept as Text rather than unnecessarily high-cardinality identifiers, keeping dictionary encoding efficient.

12. Data Dictionary Appendix

Consolidated reference of every column across the data model, for quick lookup.

Customers

Column	Type	Description
CustomerID	Text/Key	Unique customer identifier
CustomerName	Text	Full customer name
Age	Whole Number	Customer age
Gender	Text	Male / Female classification
City	Text	Customer city
State	Text	Customer state
JoinDate	Date	Registration date

Orders

Column	Type	Description
OrderID	Text/Key	Unique order identifier
CustomerID	Text/Key	Foreign key to Customers
OrderDate	Date	Date the order was placed
Region	Text	Sales region
PaymentMode	Text	Payment method used

Order_Details

Column	Type	Description
OrderDetailID	Text/Key	Unique line-item identifier
OrderID	Text/Key	Foreign key to Orders
ProductID	Text/Key	Foreign key to Products
Quantity	Whole Number	Units sold
UnitPrice	Decimal	Price per unit
Cost Price	Decimal	Cost per unit
Gross Sales	Decimal	Sales before discount

Column	Type	Description
Discount	Decimal	Discount percentage
Discount Amount	Decimal	Discount value
Net Sales	Decimal	Final sales amount

Products

Column	Type	Description
ProductID	Text/Key	Unique product identifier
ProductName	Text	Product name
Category	Text	Product category
SubCategory	Text	Product subcategory
UnitPrice	Decimal	Selling price
CostPrice	Decimal	Product cost

Calendar

Column	Type	Description
Date	Date/Key	Calendar date; joins to Orders[OrderDate]
Day	Whole Number	Day of month
Month No	Whole Number	Numeric month for sorting
Quarter	Text	Calendar quarter
Year	Whole Number	Calendar year

13. Glossary of Key Terms

A quick-reference glossary of the Power BI and data modeling terminology used throughout this document — useful both for stakeholders unfamiliar with BI tooling and as a refresher before technical interviews.

Term	Definition
Star Schema	A data model where a central fact table connects to surrounding dimension tables, each joined by a single key, minimizing join complexity.
Fact Table	A table that stores measurable, transactional events — here, Order_Details, which holds one row per line item sold.
Dimension Table	A table that stores descriptive attributes used to slice and filter facts — e.g., Customers, Products, Calendar.
Cardinality	The uniqueness of values in a relationship column; One-to-Many means one dimension row can relate to many fact rows.
Cross-Filter Direction	The direction filters propagate across a relationship — Single means filters flow from dimension to fact only.
Filter Context	The set of filters (from slicers, visuals, and relationships) applied when a DAX measure is evaluated.
Row Context	The context that exists when DAX iterates row-by-row, such as inside SUMX or a calculated column.
Context Transition	The conversion of row context into filter context, triggered by CALCULATE — the mechanism behind most DAX measures.
DISTINCTCOUNT	A DAX function that counts unique values in a column, used here to avoid double-counting orders or customers.
DIVIDE()	A DAX function for safe division that returns BLANK() (or a specified alternate value) instead of an error on division by zero.
Time Intelligence	A family of DAX functions (TOTALYTD, SAMEPERIODLASTYEAR, etc.) that calculate metrics over standard calendar periods.
Power Query	Power BI's ETL engine, used for data cleansing, shaping, and type correction before data reaches the model.
VertiPaq	Power BI's in-memory, columnar storage engine,

Term	Definition
	which is why star schemas with low-cardinality columns perform best.
Slicer	An interactive filter visual that lets end users narrow the report to a specific value or range without editing the model.

14. Assumptions, Risks & Limitations

14.1 Assumptions Log

Assumption	Rationale
Source extracts are refreshed on a scheduled (non-real-time) basis	Matches the current BRD scope; live-connection was explicitly out of scope
Currency values are in a single, consistent unit across all records	No multi-currency fields were present in the source tables
Each OrderID maps to exactly one Region and one PaymentMode	Confirmed during Power Query validation of the Orders table
Historical data volume remains within Power BI Desktop's in-memory capacity	No incremental refresh or DirectQuery was configured in this phase

14.2 Known Limitations

- No row-level security (RLS) has been configured; all users currently see the full dataset.
- Time Intelligence measures (YTD, YoY) are documented as a recommendation but not yet implemented, since the Calendar table has not been marked as an official date table.
- The model does not yet account for returns, cancellations, or refunds as a distinct transaction type.
- Forecasting and predictive analytics are outside the scope of this descriptive-analytics phase.

14.3 Risk Register

Risk	Likelihood	Mitigation
Source data schema changes without notice	Medium	Add schema validation step in Power Query before load
Dashboard adoption stalls without stakeholder training	Medium	Short walkthrough session per stakeholder group at launch
Model performance degrades as row counts grow	Low (current volume)	Monitor with Performance Analyzer; consider aggregations if needed
Metric definitions drift if measures are duplicated outside the KPI table	Medium	Enforce single-source-of-truth policy; all new measures go in KPI Measures

15. Lessons Learned & Reflections

Building this project end-to-end — from raw tables to an executive-ready dashboard — surfaced a few practical lessons worth carrying into future BI engagements:

- Modeling decisions made early (star schema, single-direction filters) paid off later by keeping every DAX measure simple; retrofitting a clean model after building measures on a messy one is far more expensive than getting the relationships right first.
- DIVIDE() and DISTINCTCOUNT became defaults rather than exceptions — small defensive habits in DAX prevent confusing blank-card bugs during stakeholder demos.
- A centralized measures table made the field list dramatically easier for non-technical stakeholders to browse — a small structural choice with an outsized usability payoff.
- Design decisions (F-pattern layout, KPI cards above the fold) mattered as much as the underlying data model for how quickly executives could trust and use the report.
- Documenting 'why' a measure or relationship exists — not just 'what' it does — is what makes a model maintainable by someone other than its original author.

15.1 What I Would Do With More Time

- Implement full Time Intelligence (YTD/MTD/YoY) with a properly marked date table.
- Add RLS so the same dashboard could be safely distributed to regional managers, restricted to their own region.
- Layer in a customer segmentation (RFM) view for the Marketing stakeholder.
- Set up a scheduled refresh against a live database rather than static extracts.

16. Project Deliverables & Conclusion

13.1 Summary of Deliverables

- Interactive Power BI Dashboard covering sales, customer, product, regional, and payment analysis
- Fully documented Star Schema data model
- Centralized DAX measure library (5 core measures, extensible)
- Executive-ready single-page report with cross-filtered visuals
- Business insights and data-backed recommendations
- Complete technical and data-dictionary documentation

13.2 Closing Statement

This engagement demonstrates the complete Business Intelligence development lifecycle: understanding a business problem, gathering requirements, designing a scalable data model, building a governed measure library, and translating raw transactional data into an executive-ready decision-support tool. The result is a single, trusted source of truth that reduces manual reporting effort and gives stakeholders the ability to explore performance across regions, categories, customers, and payment channels in real time.

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